# Corrected Greedy — new results for the paper's tables and figures

Generated from stored runs by `src/output_analysis/greedy_corrected_report.py` and `review/TRD_R1/greedy_corrected_summary.py`. The submitted Greedy runs, tables and figures are unchanged; every new file carries the suffix `_greedyC`.

## What the correction changes

> The corrected Greedy (`src/methods/greedy_corrected.py`, runs labelled `GREEDY_CORR`) keeps every decision rule of the submitted Greedy — stop only when forced, charge to full when stopping, 95%-quantile one-step guard, 10% battery buffer, queue avoidance — but executes each stop exactly as the MILP does:
>
> - **C1** A charge before a daily rest is sequential (Eq. 14): the 5-min repositioning is paid and the charge is working time.
> - **C2** A charge is credited as break only if it lasts at least the break (Eq. 15); otherwise the full break follows the charge, with the repositioning. The truck no longer idles at the bay.
> - **C3** No charge shorter than the 15-min minimum (Eq. 9).
> - **D1** At a mandatory break, a charge is added only if it is free (covers the break) or forced by the energy state.
> - **D2** The spread rules use the exact dwell the simulator executes (incl. repositioning and access time).

## Baseline vs corrected Greedy, base case (900 instances)

| Route | Infeasible old / new (%) | Gap old / new (%) | TW hit, windowed old / new (%) | Directive ok old / new (%) | Δ duration, paired (%) | Sequential stops per run, new (rest / break) |
|---|---|---|---|---|---|---|
| Short | 1.7 / 1.7 | 7.82 / 7.88 | 70.7 / 70.6 | 89.8 / 89.8 | +0.05 (n=295) | 0.16 / 0.00 |
| Medium | 8.0 / 8.0 | 7.51 / 7.57 | 54.8 / 55.1 | 90.6 / 90.9 | +0.06 (n=276) | 0.44 / 0.00 |
| Long | 3.7 / 3.7 | 6.74 / 6.99 | 58.5 / 58.2 | 92.7 / 92.7 | +0.23 (n=289) | 0.96 / 0.00 |
| All | 4.4 / 4.4 | 7.36 / 7.48 | 61.5 / 61.4 | 91.0 / 91.2 | +0.11 (n=860) | 0.52 / 0.00 |

## Table 8 — gap to oracle (%), window penalties included

| Route | Cust. | RO | 2SP | Greedy (submitted) | Greedy (corrected) | LA | Δ Greedy |
|---|---|---|---|---|---|---|---|
| Short | Few | 22.7 | 2.3 | 6.7 | 6.8 | 2.5 | +0.10 |
| | Medium | 27.7 | 3.2 | 8.0 | 8.0 | 3.6 | +0.03 |
| | Many | 34.3 | 3.4 | 8.9 | 8.9 | 2.8 | +0.04 |
| Medium | Few | 34.9 | 5.1 | 6.5 | 6.6 | 3.5 | +0.09 |
| | Medium | 33.5 | 4.5 | 6.8 | 6.8 | 2.4 | +0.07 |
| | Many | 33.9 | 6.9 | 9.4 | 9.4 | 4.0 | -0.00 |
| Long | Few | 35.7 | 10.0 | 6.1 | 6.3 | 4.4 | +0.23 |
| | Medium | 35.5 | 10.4 | 6.3 | 6.5 | 3.7 | +0.19 |
| | Many | 35.4 | 14.6 | 7.8 | 8.1 | 5.3 | +0.31 |

Means over each method's feasible runs, time-window classes pooled. All non-Greedy columns are identical to the current Table 8.

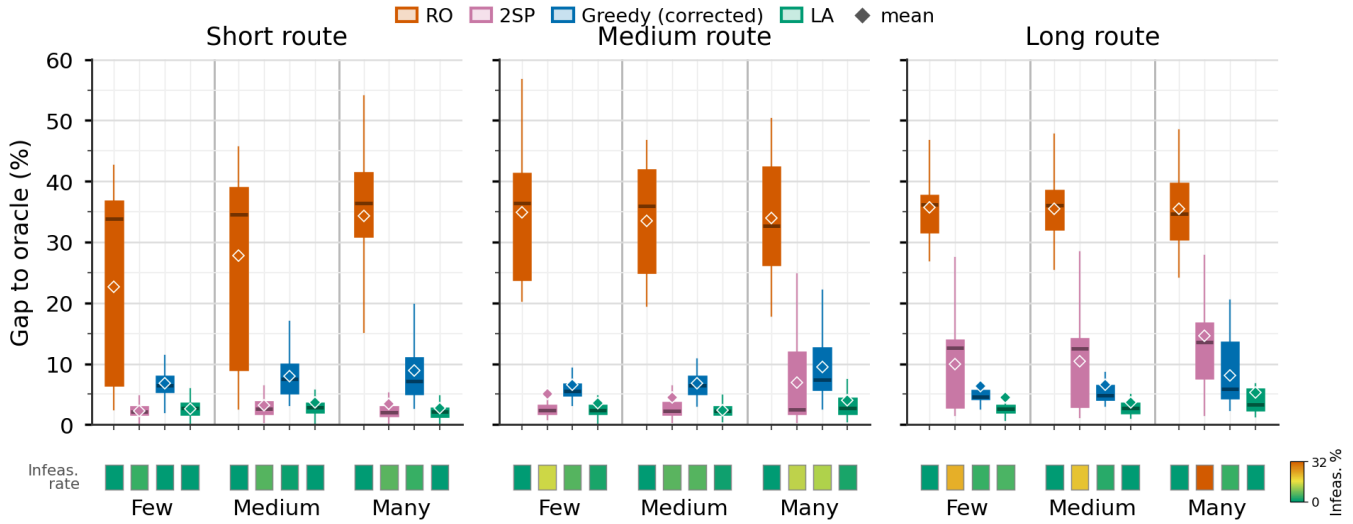
## Table 9 — feasibility (base case only)

| Method | Infeasible (%) | 95% CI | HoS | Reserve | TW hit (%) | TW hit, windowed (%) | Not opt. (%) | Opt. gap (%) |
|---|---|---|---|---|---|---|---|---|
| Greedy (corrected) | 4.4 | 3.3, 6.0 | 4.4 | 0.0 | 72.7 | 61.4 | -- | -- |
| Greedy (submitted) | 4.4 | 3.3, 6.0 | 4.4 | 0.0 | 72.1 | 61.5 | -- | -- |
| RO | 0.0 | 0.0, 0.4 | 0.0 | 0.0 | 40.2 | 21.4 | 37.4 | 4.2 |
| LA | 1.2 | 0.6, 2.2 | 0.4 | 0.4 | 91.1 | 86.1 | -- | -- |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2SP | 13.7 | 11.6, 16.1 | 0.2 | 13.5 | 83.4 | 78.5 | 54.8 | 12.1 |

Base-case runs only; the submitted Table 9 pooled the sensitivity-axis runs into its Greedy row (8.9%).

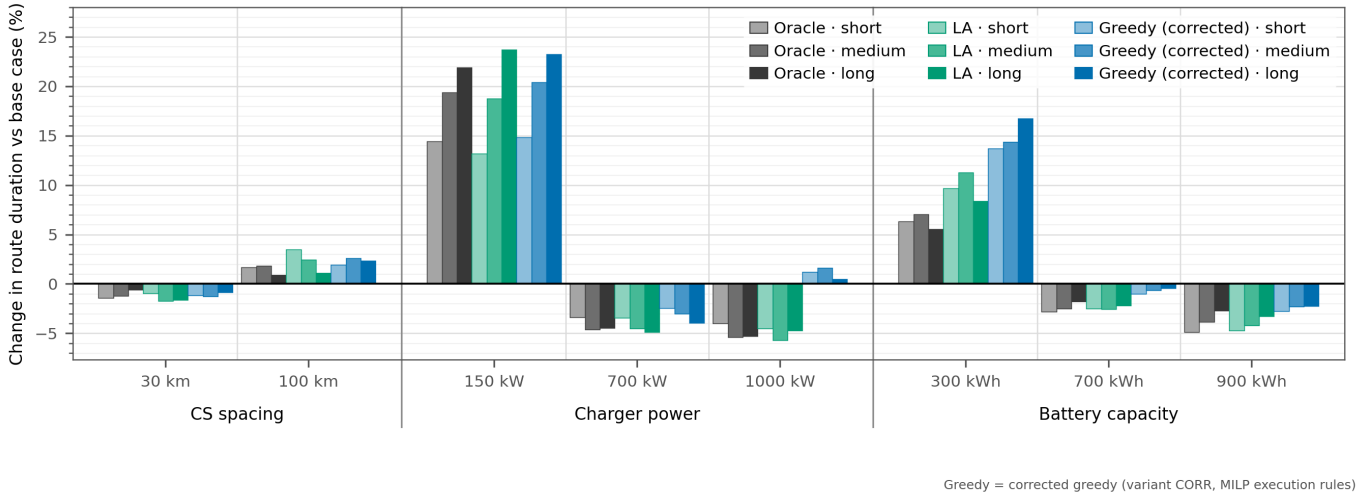
## Figure 6 — corrected Greedy




Gap to oracle by route class and customer count, time-window classes pooled; Greedy = corrected Greedy.

## Figure 9 — one-at-a-time sensitivity

| Axis level | Oracle | LA | Greedy submitted (n) | Greedy corrected (n) |
|---|---|---|---|---|
| CS spacing 30 km | -1.10 | -1.37 | -1.15 (274) | -1.09 (273) |
| CS spacing 100 km | +1.46 | +2.64 | +2.36 (261) | +2.29 (261) |
| Charger power 150 kW | +18.57 | +16.80 | +19.69 (264) | +19.61 (264) |
| Charger power 700 kW | -4.16 | -4.08 | -3.21 (272) | -3.12 (272) |
| Charger power 1000 kW | -4.89 | -5.17 | +0.34 (255) | +1.11 (259) |
| Battery 300 kWh | +6.32 | +10.05 | +14.98 (252) | +15.02 (251) |
| Battery 700 kWh | -2.36 | -2.48 | -0.62 (251) | -0.71 (257) |
| Battery 900 kWh | -3.82 | -4.30 | -2.40 (263) | -2.46 (264) |




Greedy = corrected greedy (variant CORR, MILP execution rules)

Paired change in mean route duration vs the base case; Greedy bars = corrected Greedy. Oracle unchanged. LA unchanged except the 30-km bar, which now uses the 30-km LA runs completed on 2 Oct 2026 (n = 209; the existing figure, made on 27 Aug, had 16 pairs).

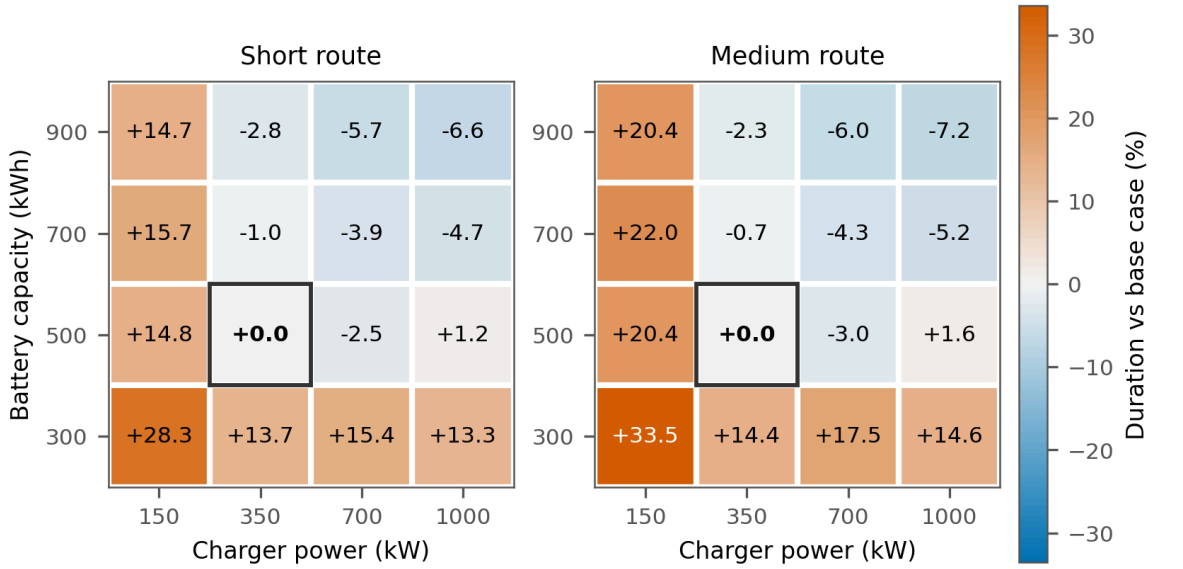
## Figure 10 — battery × power grid

### Short routes — change in Greedy route duration vs base (%), submitted → corrected

| Battery \ power | 150 kW | 350 kW | 700 kW | 1000 kW |
|---|---|---|---|---|
| 900 kWh | +14.8 → **+14.7** | -2.7 → **-2.8** | -5.8 → **-5.7** | -6.7 → **-6.6** |
| 700 kWh | +15.8 → **+15.7** | -0.9 → **-1.0** | -3.9 → **-3.9** | -4.7 → **-4.7** |
| 500 kWh | +14.9 → **+14.8** | +0.0 → **+0.0** | -2.6 → **-2.5** | +0.5 → **+1.2** |
| 300 kWh | +28.1 → **+28.3** | +13.6 → **+13.7** | +13.3 → **+15.4** | +11.9 → **+13.3** |

### Medium routes — change in Greedy route duration vs base (%), submitted → corrected

| Battery \ power | 150 kW | 350 kW | 700 kW | 1000 kW |
|---|---|---|---|---|
| 900 kWh | +20.6 → **+20.4** | -2.3 → **-2.3** | -6.3 → **-6.0** | -7.2 → **-7.2** |
| 700 kWh | +21.9 → **+22.0** | -0.6 → **-0.7** | -4.2 → **-4.3** | -5.2 → **-5.2** |
| 500 kWh | +20.4 → **+20.4** | +0.0 → **+0.0** | -3.3 → **-3.0** | +0.4 → **+1.6** |
| 300 kWh | +33.3 → **+33.5** | +14.3 → **+14.4** | +14.9 → **+17.5** | +13.1 → **+14.6** |

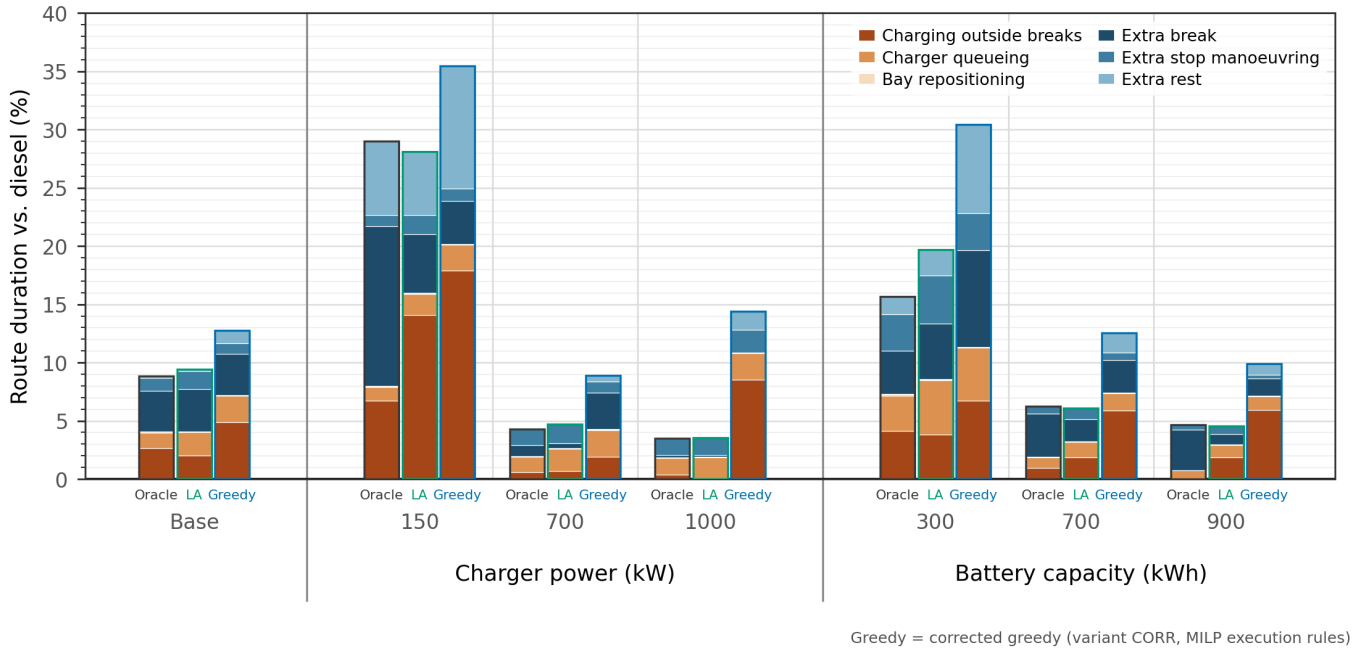



Greedy = corrected greedy (variant CORR, MILP execution rules)

Corrected Greedy route duration against the base case.

## Figure 11 — electrification penalty vs diesel (%)

| Level | Oracle | LA | Greedy submitted (n) | Greedy corrected (n) |
|---|---|---|---|---|
| Base | 8.8 | 9.4 | 12.6 (267) | **12.7** (268) |
| 150 kW | 29.0 | 28.1 | 35.4 (263) | **35.5** (263) |
| 700 kW | 4.3 | 4.7 | 8.7 (269) | **8.9** (269) |
| 1000 kW | 3.5 | 3.5 | 13.3 (253) | **14.4** (259) |
| 300 kWh | 15.7 | 19.7 | 30.2 (250) | **30.4** (250) |
| 700 kWh | 6.2 | 6.1 | 12.3 (248) | **12.5** (256) |
| 900 kWh | 4.6 | 4.6 | 9.7 (262) | **9.9** (264) |




Greedy = corrected greedy (variant CORR, MILP execution rules)

Each EV method against its diesel counterpart; Greedy = corrected Greedy on both legs.